Practice Problem on hand simulating caches

Consider the following trace of addresses:

1. 0xA1D4
2. 0x31A7
3. 0xA1E6
4. 0xFF44
5. 0xFF58
6. 0x112D
7. 0x1132
8. 0xC1D0
9. 0xFF78
10. 0x31D2
11. 0x1110
12. 0x1126
13. 0x10D0
14. 0x01E2
15. 0xFF78
16. 0x113A
17. 0xA1D6
18. 0xA1DA
19. 0x1126
20. 0x10D5

In this question, assume that the cache uses an LRU eviction policy. Hand simulate the cache accesses for the following cache dimensions and indicate if an access is a miss. If an access is a miss, indicate the type of miss.

a) (C= 7, B =4, S=0)
b) (C= 7, B =4, S=3)
c) (C= 7, B =4, S=1)